Homework 26: 10.4-10.6 **Name** _____

AP Calculus BC **Due** _____

Period _____ **Score** _____ **/ 100**

Skill Builder 10.4 – The Integral Test for Convergence

In the following exercises, use the Integral Test to determine the convergence or divergence of the series. Be sure to properly state whether the conditions of the test apply and justify your conclusion.

1.) $\sum_{n=1}^{\infty} \frac{1}{n+5}$
2.) $\sum_{n=0}^{\infty} e^{-n}$
3.) $\sum_{n=0}^{\infty} ne^{-n}$
4.) $\sum_{n=1}^{\infty} \frac{1}{5n+1}$

5.) $\frac{4}{2}+\frac{4}{5}+\frac{4}{10}+\frac{4}{17}+\frac{4}{26}+\cdots$

6.) $\frac{1}{5}+\frac{1}{7}+\frac{1}{9}+\frac{1}{11}+\cdots$

7.) $\frac{1}{3}+\frac{2}{6}+\frac{3}{11}+\cdots+\frac{n}{n^2+2}+\cdots$

8.) $\frac{\ln 20}{20}+\frac{\ln 21}{21}+\frac{\ln 22}{22}+\frac{\ln 23}{23}+\cdots$

9.) $\sum_{n=1}^{\infty}\frac{1}{n^4}$

10.) Find the sum of the series

$$\sum_{n=1}^{\infty}\left(\frac{2}{5^n}+\frac{3}{(n+2)(n+4)}\right)$$

Skill Builder 10.5 – Harmonic Series and *p*-Series

For the following *p*-series, determine whether they are convergent or divergent.

1.) $\sum_{n=1}^{\infty} \frac{1}{\sqrt[4]{n}}$	**2.)** $\sum_{n=1}^{\infty} \frac{1}{n^{5/4}}$
3.) $\sum_{n=1}^{\infty} \frac{1}{n^{\pi}}$	**4.)** $\sqrt{3} + \frac{\sqrt{3}}{\sqrt{2}} + \frac{\sqrt{3}}{\sqrt{3}} + \frac{\sqrt{3}}{\sqrt{4}} + \cdots$
5.) $1 + \frac{1}{2\sqrt[3]{2}} + \frac{1}{3\sqrt[3]{3}} + \frac{1}{4\sqrt[3]{4}} + \cdots$	**6.)** $1 + \frac{1}{\sqrt[5]{4}} + \frac{1}{\sqrt[5]{9}} + \frac{1}{\sqrt[5]{16}} + \cdots$

Problems 7 and 8.

For a convergent p-series, although it is not possible to find its sum, it is possible to find a lower and an upper bound for the sum. For $p > 1$, the bounds for the sum of a convergent p-series is defined by

$$\frac{1}{p-1} < \sum_{k=1}^{\infty} \frac{1}{k^p} < 1 + \frac{1}{p-1}.$$

While you will not need to know this for the AP Exam, many college Calculus 2 courses mention this inequality.

7.) Find the lower and upper bound for $\sum_{k=1}^{\infty} \frac{1}{k^{5/4}}$	**8.)** Find the lower and upper bound for $\sum_{k=1}^{\infty} \frac{3}{k^{4/3}}$

<u>Review:</u> Determine the convergence or divergence of the series and specify what test you are using.

9.) $\sum_{n=1}^{\infty} \frac{1}{3n-2}$	**10.)** $\sum_{n=1}^{\infty} \frac{2n}{1+n^4}$
11.) $4\sum_{n=1}^{\infty} \frac{1}{n^{0.97}}$	**12.)** $\sum_{n=0}^{\infty} \left(\frac{7}{8}\right)^n$

13.) $\sum_{n=1}^{\infty}\left(\frac{1}{n^2}-\frac{1}{n^3}\right)$

14.) $\sum_{n=1}^{\infty}\frac{n}{e^n}$

15.) $\sum_{n=0}^{\infty}\frac{\tan^{-1} n}{n^2+1}$

16.) $\sum_{n=2}^{\infty}\frac{n}{\sqrt{n^2-1}}$

17.) Find the sum of the series $\sum_{n=1}^{\infty}\left(\frac{3^{n-1}+2^{n-1}}{4^{n-1}}\right)$

Skill Builder 10.6 – Comparison Tests for Convergence

Use the Direct Comparison Test to determine the convergence or divergence of each of the following series.

1.) $\sum_{n=1}^{\infty} \frac{1}{n^2+2}$	**2.)** $\sum_{n=1}^{\infty} \frac{1}{3n^2+2}$
3.) $\sum_{n=4}^{\infty} \frac{1}{n-3}$	**4.)** $\sum_{n=5}^{\infty} \frac{1}{\sqrt{n}-2}$
5.) $\sum_{n=0}^{\infty} \frac{1}{4^n+1}$	**6.)** $\sum_{n=0}^{\infty} \frac{2^n}{4^n+1}$

7.) $\sum_{n=1}^{\infty} \frac{\ln n}{n+2}$	**8.)** $\sum_{n=1}^{\infty} \frac{1}{n!}$
9.) $\sum_{n=0}^{\infty} e^{-n^2}$	

Use the Limit Comparison Test to determine the convergence or divergence of each of the following series.

10.) $\sum_{n=1}^{\infty} \frac{n}{n^2+2}$	**11.)** $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n^2-2}}$
12.) $\sum_{n=1}^{\infty} \frac{1}{2^n-1}$	**13.)** $\sum_{n=1}^{\infty} \frac{3n^2-4}{4n^2+3n-1}$

14.) $\sum_{n=1}^{\infty} \frac{3n-2}{n^2+n-1}$	**15.)** $\sum_{n=1}^{\infty} \sin\frac{1}{n}$

<u>Review:</u> For the following problems, test for convergence or divergence using each test at least once. Identify the test used and show your work.

a) nth Term Test **b)** p-Series Test **c)** Geometric Series Test **d)** Integral Test

e) Telescoping Series Test **f)** Direct Comparison Test **g)** Limit Comparison Test

16.) $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n}$	**17.)** $\sum_{n=0}^{\infty} 4\left(\frac{1}{4}\right)^n$
18.) $4\sum_{n=1}^{\infty} \frac{2}{3^n+1}$	**19.)** $\sum_{n=1}^{\infty} \frac{1}{4n^2-2n+1}$

20.) $\sum_{n=1}^{\infty} \frac{n}{2n+5}$	**21.)** $\sum_{n=1}^{\infty} \left(\frac{1}{n+2} - \frac{1}{n+3} \right)$
22.) $\sum_{n=1}^{\infty} \frac{n}{\left(n^2+1\right)^2}$	**23.)** $\sum_{n=1}^{\infty} \frac{4}{n(n+4)}$